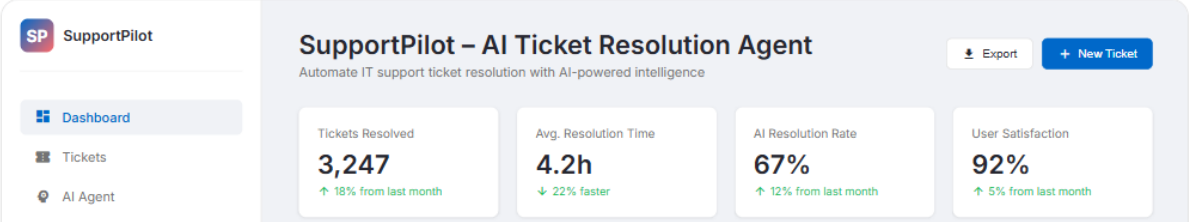
1. Title

SupportPilot – AI Ticket Resolution Agent



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2. Project Statement

Modern organizations receive a large volume of IT support requests related to network connectivity, software installation, password resets, VPN access, hardware issues, and system troubleshooting. Support teams spend significant time resolving repetitive tickets, resulting in increased operational costs, delayed response times, and reduced productivity. This project aims to develop SupportPilot, an AI-powered Ticket Resolution Agent that automates ticket classification, knowledge retrieval, solution generation, and resolution workflows. By leveraging Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), multi-agent AI architectures, and enterprise integrations, the system can analyze incoming support tickets, retrieve relevant troubleshooting information, generate contextual responses, and escalate unresolved issues when necessary. The solution improves support efficiency, reduces ticket resolution time, and enhances employee satisfaction.

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3. Outcomes

• Automated ticket intake and categorization for IT support requests.

• Intelligent severity and priority classification of support tickets.

• Context-aware troubleshooting using enterprise knowledge bases.

• AI-generated support responses and resolution recommendations.

• Automated ticket routing and escalation workflows.

• Multi-agent collaboration for complex issue investigation.

• Jira integration for ticket tracking and lifecycle management.

• Email automation for user communication and status updates.

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4. Modules to be Implemented

1. Ticket Intake & Classification

• Receive support tickets from web forms, email, and ticketing systems.

• Categorize tickets based on issue type and business impact.

• Assign severity and priority levels automatically.

2. Knowledge Retrieval Engine

• Search enterprise knowledge bases and troubleshooting guides.

• Retrieve relevant support documentation.

• Support semantic search using vector databases.

3. AI Resolution Generator

• Generate contextual troubleshooting recommendations.

• Provide step-by-step resolution workflows.

• Draft automated responses for end users.

4. Multi-Agent Resolution Framework

• Utilize specialized agents for diagnosis, retrieval, and response generation.

• Coordinate multiple agents for complex support scenarios.

• Automate resolution validation and confidence scoring.

5. Escalation & Workflow Management

• Escalate unresolved tickets to human support teams.

• Route tickets to appropriate departments.

• Track ticket lifecycle and resolution progress.

6. Dashboard & Analytics

• Interactive dashboard for support operations monitoring.

• Display ticket trends, resolution metrics, and agent performance.

• Generate support analytics and service reports.

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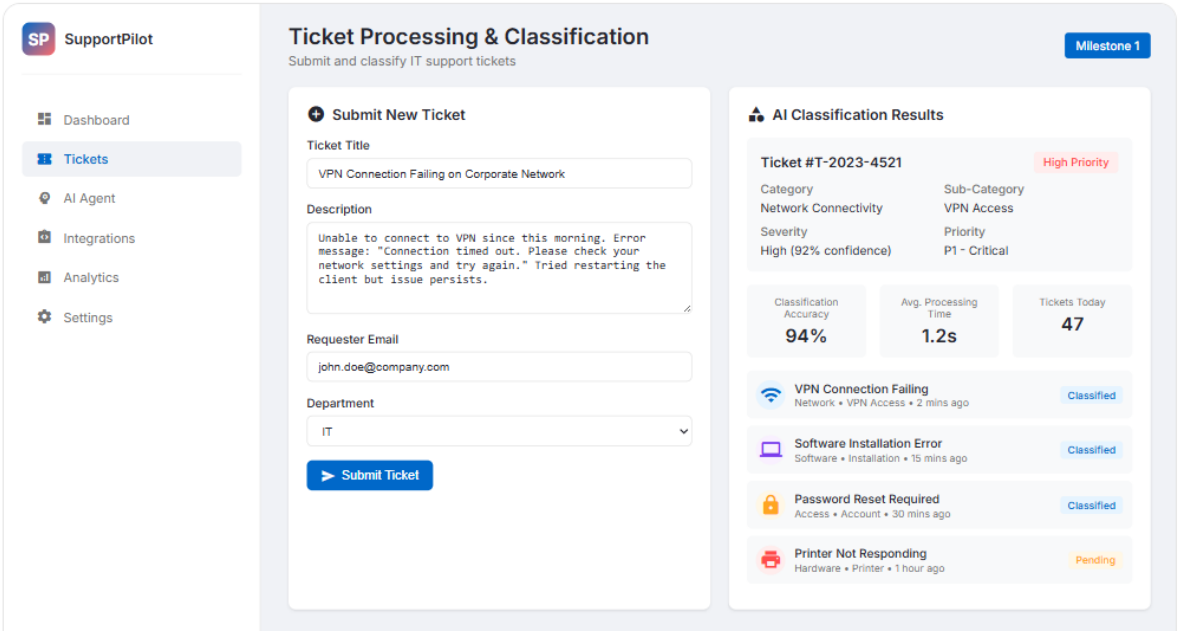
5. Week-wise Module Implementation & High-Level Requirements

Milestone 1 (Weeks 1–2) – Ticket Processing & Classification

• Implement ticket submission and ingestion functionality.

• Develop AI-based ticket classification engine.

• Generate severity and priority scores.

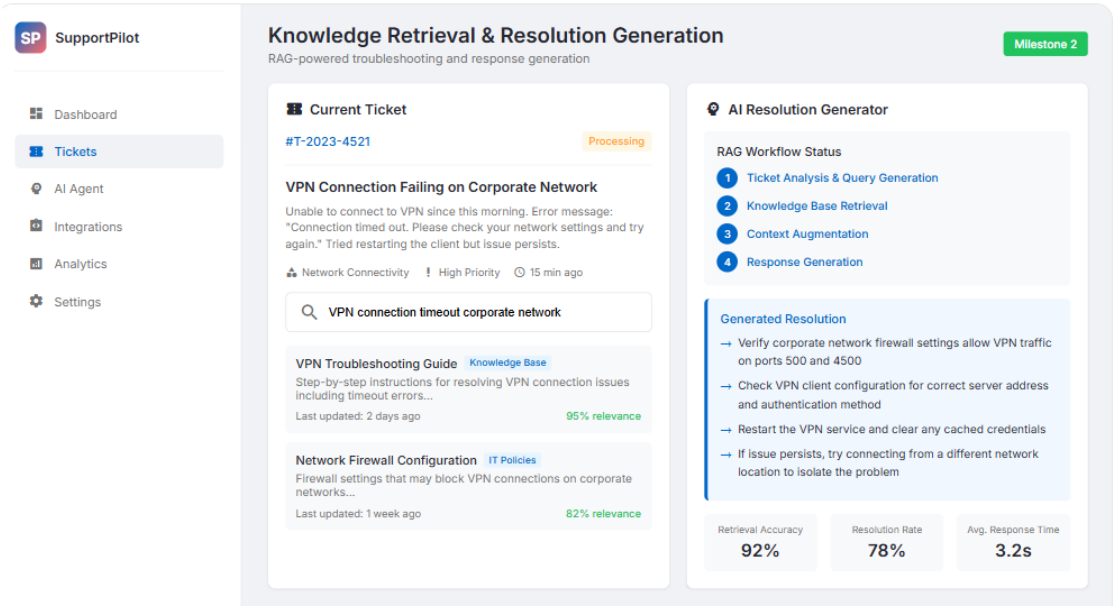


Milestone 2 (Weeks 3–4) – Knowledge Retrieval & Resolution Generation

• Integrate enterprise knowledge base search.

• Implement Retrieval-Augmented Generation workflows.

• Generate automated troubleshooting responses.

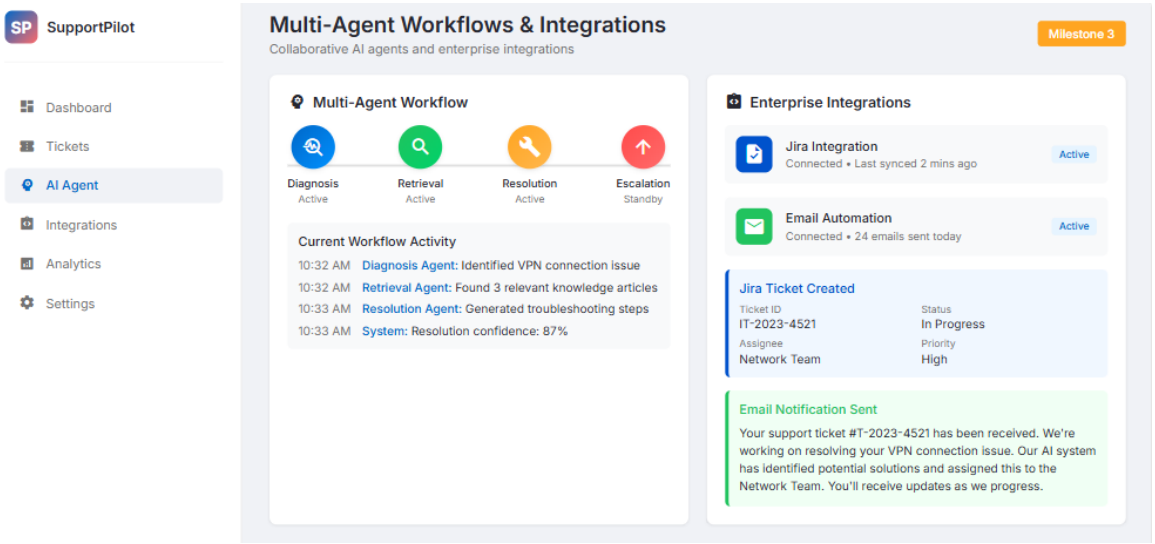


Milestone 3 (Weeks 5–6) – Multi-Agent Workflows & Integrations

• Develop multi-agent architecture for issue resolution.

• Integrate Jira for ticket management.

• Implement email automation for support communications.

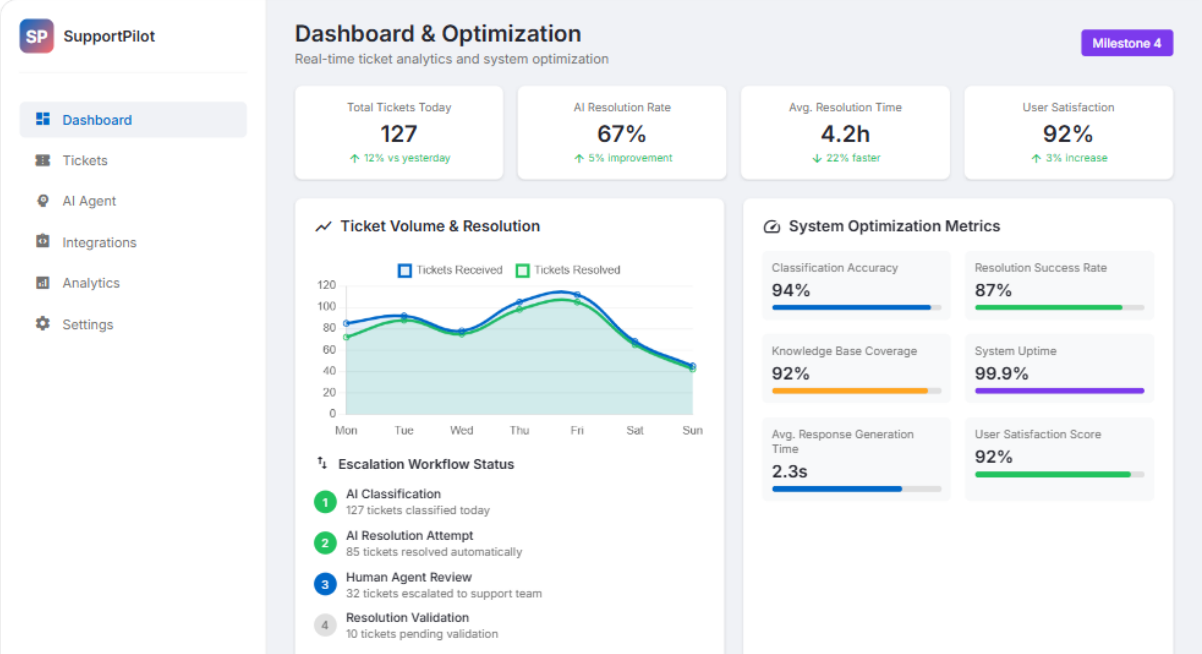


Milestone 4 (Weeks 7–8) – Dashboard & Optimization

• Develop support analytics dashboard.

• Implement escalation workflows and monitoring.

• Perform testing, optimization, and deployment.



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6. Evaluation Criteria

Milestone 1 (Week 2)

• Successfully classify ≥90% of support tickets.

• Severity prediction accuracy ≥85%.

Milestone 2 (Week 4)

• Retrieve relevant knowledge base articles for support requests.

• Generate accurate troubleshooting responses with source references.

Milestone 3 (Week 6)

• Multi-agent workflows execute successfully for complex issues.

• Jira integration functions correctly for ticket management.

• Email automation delivers support communications successfully.

Milestone 4 (Week 8)

• Dashboard fully operational with real-time ticket analytics.

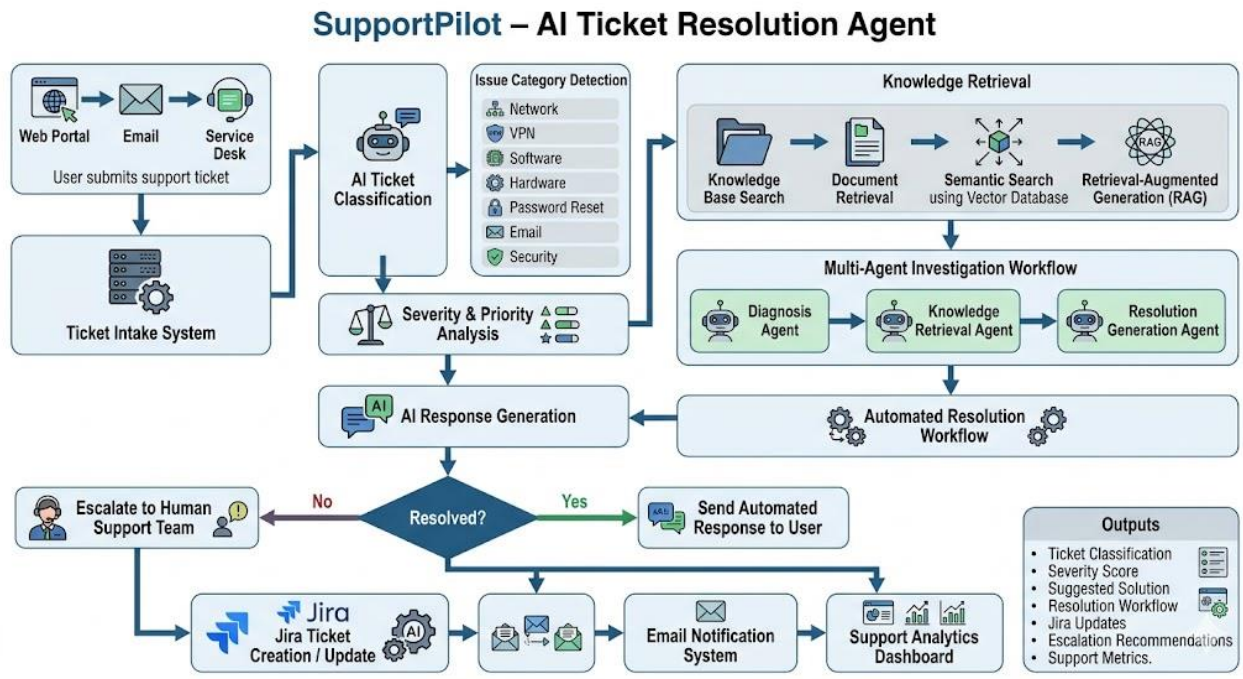
• End-to-end ticket resolution workflow operational.

• Average response generation time <5 seconds.

• User satisfaction score ≥85% during testing and evaluation.

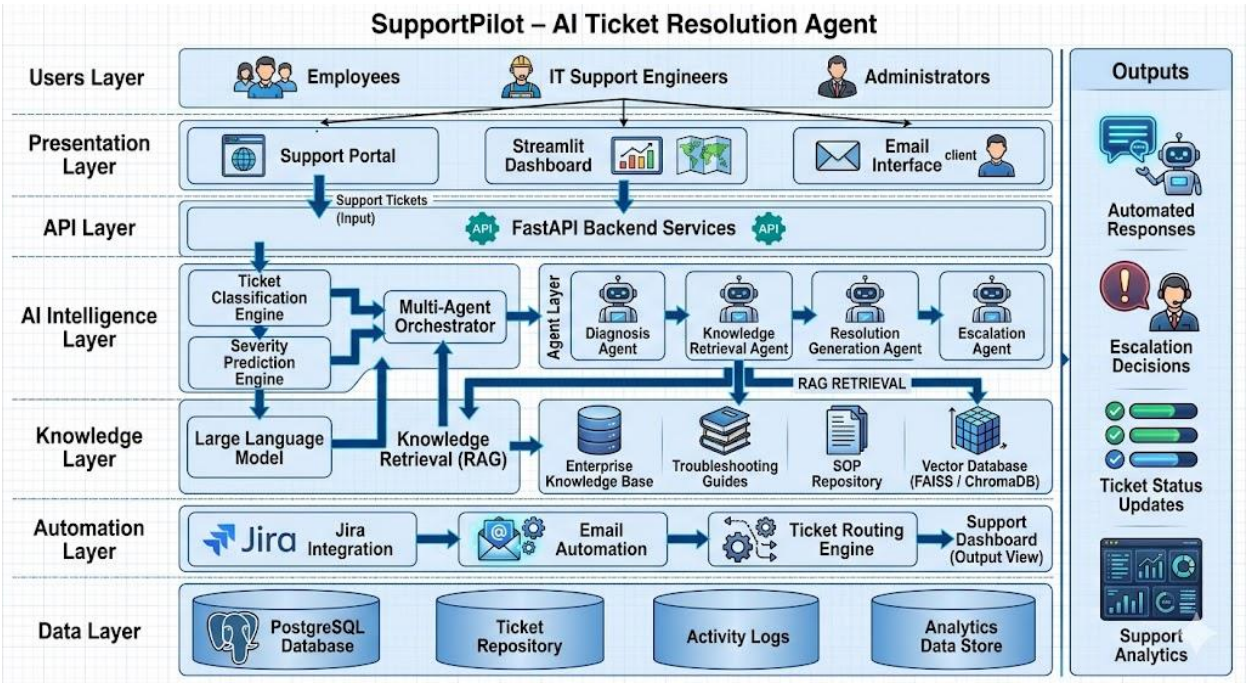
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7. Workflow Diagram



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8. Architecture Diagram



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9. Database Schema

